PSS®E-35 Release Notes



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Chapter 1
Updated Branch Loading Calculation

1.1. Changes in the Branch Loading Calculation in PSS®E-35

Branch loading calculation is performed in PSS®E with activities RATE, OLTL and OLTR. The loading in network branches (transmission lines and power transformers) is calculated as MVA loading in percent of the selected branch thermal rating in MVA or as current loading in MVA units in percent of the selected branch thermal in MVA.

In PSS®E-34 and earlier versions of PSS®E, the current loading in units of MVA for both transmission lines and transformers is calculated with the following expressions:

$$I_{flow} = \frac{MVA_{flow}}{\sqrt{3} \times kV_{actual}} \times 1000 \quad (A) \tag{1.1}$$

$$I_{rating} = \frac{MVA_{rating}}{\sqrt{3} \times kV_{rating}} \times 1000 \quad (A) \tag{1.2}$$

The terms kV_{actual} and kV_{rating} represents the bus voltage magnitude in kV from the load flow solution and the bus voltage base, kVbase, in kV, respectively. From these expressions the current loading in units of MVA is obtained:

$$\frac{I_{flow}}{I_{rating}} \times 100 = \frac{MVA_{flow}}{MVA_{rating} \times \frac{kV_{actual}}{kV_{rating}}} = \frac{MVA_{flow}}{MVA_{rating} \times V_{pu}} \times 100 \tag{1.3}$$

Note that for transformers this is correct as long as the windings rated voltage (NOMV in PSS®E transformer data record) matches the voltage base of the bus to which it is connected; however, when the a winding rated voltage, kV_{nom}, does not match its corresponding bus voltage base, kV_{base}, in equation 1.2 and 1.3 should be adjusted as follows:

$$I_{rating} = \frac{MVA_{rating}}{\sqrt{3} \times kV_{nom}} \times 1000 \quad (A) \tag{1.4}$$

$$\frac{I_{flow}}{I_{rating}} \times 100 = \frac{MVA_{flow}}{MVA_{rating} \times \frac{kV_{actual}}{kV_{base}}} \times \frac{kV_{nom}}{kV_{base}} = \frac{MVA_{flow}}{MVA_{rating} \times V_{pu}} \times \frac{kV_{nom}}{kV_{base}} \times 100 \tag{1.5}$$

Equation 1.5 has been implemented in PSS®E-35.0 and thus, for those transformers with winding rated voltage different from its corresponding bus voltage base and with entries NOMV different from its default value of 0.0, will show a current loading in units of MVA different from that obtained with earlier versions of PSS®E. Note that in PSS®E if the entries in the transformer data record NOMV are their default value (0.0), PSS®E will assume $kV_{nom} = kV_{base}$ and hence equation 1.3 will be used.

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Chapter 2
Python Compatibility

2.1. Python 2.7 End of Life

Python 2.7 reached end of life on January 1st, 2020 and is no longer being maintained. We strongly recommend users update their Python scripts used with PSS®E to be compatible with Python 3.7. PSS®E-34 has introduced full support for Python 3.7 as of PSS®E-34.7 while PSS®E-35.0 has standardized on using Python 3.7 by default. While PSS®E-34 and PSS®E-35 will continue to provide Python 2.7 support, this may change in the future due to maintenance and/or security concerns.

PSS®E automation Python scripts will continue to work in Python 3.x without any changes. However, certain user developed Python scripts may need minor modifications during the migration process. Several common areas of attention are included below.

To assist in transitioning Python 2.7 scripts to Python 3, users can use the "2to3" utility provided with the Python 3.7 installation package. As an example, if "test.py" is a Python 2.x script, it can be converted to a Python 3.x script [test.py3] by running the following command at the Command Prompt:

```
C:\Python37\python.exe C:\Python37\Tools\scripts\2to3.py -n -W --add-
suffix=3 "test.py"
```

[Here it is assumed that Python 3.7 is installed in folder C:\Python37 and this command is run from the directory where test.py is located.]

Additional information can be found on python.org at https://docs.python.org/3/howto/pyporting.html and https://docs.python.org/3/whatsnew/3.0.html.

Common areas that should be updated for compatibility include (but are not limited to):

- print has been changed to print()
- Integer division
- execfile() behavior changed to exec(open(file).read())
- raw_input() renamed to input()

2.2. Python 3.8

Support for Python 3.8 is being added for the PSS®E-35.2.0 release. The following summarizes what this means:

- PSS®E modules compatible with Python 3.8 will be installed with PSS®E-35.2.0.
- The default version of Python used with PSS®E-35.2.0 (i.e. the version that is used by PSS®E interfaces without specifying a version with -pyver) will not change. It will continue to be Python 3.7 as it has been for PSS®E-35 earlier releases.

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- A Python 3.8 installer will be installed, but not run by the PSS®E-35.2.0 installer (the default location for the Python 3.8 installer will be C:\Program Files\PTI\PSS®E35 Installer Files\35.2\Python\Python38x64).
- The modules psse35, psse3502, psseloc, and psse_env_manager will be copied to your local installations of (64-bit) Python 2.7, Python 3.7, and Python 3.8 by the PSS®E-35.2.0 installer if they are present at the time.

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Chapter 3
Compatibility Matrices

3.1. PSS®SINCAL Compatibility

This section outlines compatibility between PSS®SINCAL and PSS®E versions.

Table 3.1. PSS®SINCAL Compatibility

PSS®E	BOSL in PSS®E - Using models	License needed	GMB - Editing models	License needed	License from	Eigen-value Analysis	License needed	License from	Har-monics	License needed
35.0	15.5	No	15.5	Yes	SINCAL	15.5	Yes	SINCAL	NA	-
35.0	16.0	No	16.0	Yes	SINCAL	16.0	Yes	SINCAL	NA	-
35.1	16.0	No	16.0	Yes	SINCAL	16.0	Yes	SINCAL	NA	-
35.1	16.5	No	16.5	Yes	PSS®E	16.5	Yes	PSS®E	NA	-
35.1	17.0	No	17.0	Yes	PSS®E	17.0	Yes	PSS®E	NA	-
35.2	16.5	No	16.5	Yes	PSS®E	16.5	Yes	PSS®E	NA	-
35.2	17.0	No	17.0	Yes	PSS®E	17.0	Yes	PSS®E	NA	-

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Chapter 4
Deprecated Utilities and APIs

The following utilities and APIs have been deprecated and will be removed from the PSS®E installer at a future release.

4.1. Deprecated Utilities

The following utilities have been deprecated and will be removed at PSS®E-36.0:

1. Utilities that are used to convert RAW and SEQ files between different versions:

   - CNV27
   - CNV29
   - CNV30
   - CNV31
   - CNV32
   - CNVRAW
   - CONVERTRAW
   - CNVRSQ

   The functionality of the above conversion utilities remain directly accessible from the PSS®E GUI, File>Open, which can read any current or previous versions of the RAW and SEQ files.

   These functions are also available through automation scripts: psspy.readrawversion(..), psspy.resqversion(..), bat_readrawversion and bat_resqversion.

2. Utilities that are used to write previous versions of RAW and SEQ files:

   - RAW23
   - RAW26
   - RAW28
   - RAW29
   - RAW30
   - RAW31
   - CREATERAW

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The functionality of these conversion utilities are directly accessible from the PSS®E GUI, File>Save Case As..., which can write any current or previous versions of the RAW and SEQ files.

These functions are also available through automation scripts: psspy.writerawversion(..), psspy.writeseqversion(..), bat_writerawversion and bat_writeseqversion.

3. Converting/writing power flow data, *.raw, and sequence data, *.seq, formats to versions 28 and prior is no longer supported. We will continue to support reading raw and seq files from any prior version.

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Chapter 5
PSS®E-35.2.0 Release Notes

November 2020

5.1. Discontinued Modules

The Measurement Interface and the Data Visualization and Reporting Module (DVRM) have been dropped from PSS®E, starting at PSS®E-35.1. If you are currently licensed for these modules you may continue to use these modules in your PSS®E-35.0 installation.

If you wish to continue using your Measurement Interface and DVRM modules, do *NOT* uninstall PSS®E-35.0 before installing PSS®E-35.2. Failure to preserve your PSS®E-35.0 installation will result in the loss of these modules.

5.2. Beta Functionality

The following functionality is being introduced in this release as beta functionality. That means that there still might be issues with it and that it does not support all the functionality that the final production version will. It is being offered here in the hope that you will try it out and give us your feedback.

The following beta functionality appears in this release:

1. Generators can now model voltage droop control. In voltage droop control, a QV characteristic curve is specified where the reactive power at the controlled bus is a function of the controlled bus voltage. The purpose of the curve is to enforce that the total reactive flow going into the bus from the generators that belong to the power plant control is equal to this value.

   Note: The QV characteristic curve and voltage droop control assignment are defined in the PSS®E extensible raw file (*.rawx) using extension tables fields. The voltage droop control is supported only in the Full Newton-Raphson method (FNSL).

2. The support of vendor neutral Remedial Action Schemes (RAS) is expanded to AC Contingency analysis (ACCC). ACCC uses the SUB/CON/MON file to define subsystems, contingencies and monitored elements, plus optionally the auxiliary files (commonly called AUX file) to define remedial action schemes. A new option, added to the contingency analysis parameters, is used to enable or disable the remedial action schemes in ACCC. When the option is enabled and remedial action schemes are loaded, for each contingency, ACCC will perform a complete procedure to evaluate each contingency and all remedial action schemes triggered by the contingency.

   Note: Besides remedial action schemes, AUX files may include contingency definitions and monitored elements, however when the AUX files are used by ACCC, only remedial action schemes are considered. The integration of SUB/MON/CON/AUX files is supported only in ACCC.

5.3. Program Enhancements

The following program enhancements appear in this release:

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1. Power Flow

   - wxPython scripts can now be launched from a Custom Toolbar button.

2. The following Report Tables have been added to PSS®E:

   - Preventive Security Constrained OPF (PSCOPF) report table
   - DC PSCOPF report table
   - AC corrective action report table
   - DC corrective action report table
   - RAS report table

3. RAWX Extensions in Data Record Dialogs

   RAWX extension data (user-defined fields) can now be viewed and edited in the data record dialogs via the “Extension Data” tab.

4. RAWX CSV Import/Export

   RAWX data can now be Imported/Export to comma-separated value (CSV) files via the GUI as well as the API. Use File>Import>RAWX from CSV and File>Export>RAWX to CSV, or the APIs CSV_TO_RAWX and RAWX_TO_CSV. All RAWX data is supported, built-in RAW file data as well as user extension fields. When importing from CSV, the RAWX data is treated as appending/editing the exisiting data (similar to RAW data with IC=1, or using RDCH).

5. Advanced Linear Analysis/MUST

   While the original MUST APIs continue to be supported, new APIs have been introduced to supplant the older APIs with ones that are more comprehensive. The older MUST functions have an atomic structure in which each function executes a small portion of the desired calculation. This scheme results in requiring perhaps several API calls to modify options, more calls to other APIs for initialization and finally a separate call to another API to perform the actual calculation. Sometimes, many API calls are needed to make a single calculation. While this scheme provides incremental flexibility, it can be quite cumbersome to use and requires intimate knowledge of the old MUST *internal command* structure.

   With this version of PSS®E, new APIs have been added to the Advanced Linear Analysis/MUST module which will automatically call the necessary atomic functions so the user need only worry about the desired outcome. These new, full function, APIs require, as arguments, all the necessary options relevant to that calculation as well as the required inputs to generate the desired output. They are structured to mimic the actual dialog entries encountered when using the PSS®E GUI to run an Advanced Linear Analysis/MUST calculation. That is, the inputs presented by the calculation's dialog in the GUI are exactly the inputs required by the new API. In this way, familiarity with the GUI will automatically translate into familiarity with the respective API.

   The following APIs have been added to the Advanced Linear Analysis/MUST module to assist users to generate reports programmatically:

   - ACLFNS: to execute an AC load flow solution using the Full Newton Method and specified calculation options.

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- ACLFDC: to execute an AC load flow solution using the Fast Decoupled Method and specified calculation options.
- ACCSOLVE: to run the AC contingency analysis solution with the specified calculation options.
- ILFDEFAULTS: to enter a set of default Initial Line Flow options for later use in AC load flow solutions.
- ACCDEFAULTS: to enter a set of default AC Contingency options for later use in AC contingency analysis.
- SGLDCFCITC: to calculate the effects of a Single Linear FCITC Transfer.
- MULTDCFCITC: to calculate the effects of a series of Linear FCITC Transfers.
- SGLACFCITC: to calculate the effects of a Single AC FCITC Transfer.
- MULTACFCITC: to calculate the effects of a series of AC FCITC Transfers.
- DCCONTBYCONT: to generate a DC contingency report by contingency options.
- DCCONTBYMONEL: to generate a DC contingency report by monitored elements.
- ACCONTBYCONT: to generate an AC contingency report by contingency options.
- ACCONTBYMONEL: to generate an AC contingency report by monitored elements.

6. Contingency Analysis

- The DC based Preventive Security Constrained Optimal Power Flow solution (DCPSCOPF) is provided. This API is used to find the settings of controls for a system to comply with base case and contingency case operation limits. The DCPSCOPF is super fast because it uses a DC load flow network and linear programming solution. The objective of the DCPSCOPF is to minimize control adjustments subject to operation limit constraints. The available controls are: generation MW outputs of on-line and off-line generators, loads, phase shifters.
- Added the following items to the contingency (CON) file:

  a. SET SWSHUNT AT BUS xxx REACTIVE TO xxx MVAR

  b. SET SWSHUNT AT BUS xxx REACTIVE TO xxx REACTIVE

  c. SET SWSHUNT AT BUS xxx VMAX TO xxx PERUNIT

  d. SET SWSHUNT AT BUS xxx VMIN TO xxx PERUNIT

  e. SET SWSHUNT AT BUS xxx VSCHED TO xxx PERUNIT

  f. CHANGE SWSHUNT AT BUS xxx REACTIVE BY xx MVAR

  g. CHANGE SWSHUNT AT BUS xxx REACTIVE BY xx PERCENT

  h. CHANGE SWSHUNT AT BUS xxx VMAX TO xxx PERUNIT

  i. CHANGE SWSHUNT AT BUS xxx VMIN TO xxx PERUNIT

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j. CHANGE SWSHUNT AT BUS xxx VSCHED TO xxx PERUNIT

k. MOVE SWSHUNT xxx xxx MVAR FROM BUS xxx TO BUS xxx

l. MOVE SWSHUNT xxx xxx PERCENT FROM BUS xxx TO BUS xxx

m. ADD SWSHUNT xxx TO BUS xxx

n. ISOLATE MACHINE xxx AT BUS xxx

o. ISOLATE LOAD xxx AT BUS xxx

p. ISOLATE SHUNT xxx AT BUS xxx

q. ISOLATE SWSHUNT AT BUS xxx

r. ISOLATE INDUCMACHINE xxx AT BUS xxx

s. ISOLATE TWOTERMDC xxx

t. ISOLATE MULTITERMDC xxx

u. ISOLATE VSCDC xxx

v. ISOLATE FACTS xxx

w. SINGLE ISOLATE BUS IN xxx

x. SINGLE ISOLATE BREAKER IN xxx

y. SINGLE ISOLATE SWITCH IN xxx

z. SINGLE ISOLATE SWD IN xxx

- Added the following enhancements to the monitored element (MON) file:

  a. For all subsystem specifications, these inclusive range definitions are now allowed:

     - AREAS i j
     - ZONES i j
     - OWNERS i j
     - KVRANGE r1 r2

  b. For all voltage monitoring records, this additional inclusive range definition is now allowed:

     - BUSES i j

  c. For all MONITOR BRANCHES or BREAKERS or SWITCHES or SWDS or TIES (either or both subsystems) the subsystem specification may include an additional optional clause as follows:

     - IN KVRANGE r1 r2

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d. For all voltage monitoring records the subsystem specification may include an additional optional clause as follows:

- KV r1
- KVRANGE r1 r2

e. The keyword "TIE" may be used as a synonym for "TIES".

- Added the following commands to all of the contingency specification files, i.e. subsystem definition (SUB), monitored element (MON), contingency definition (CON), tripping element (TRP), reliability outage statistics (PRB).

  - BUSNAMES
  - BUSNUMBERS

  These can be used to dynamically switch between bus numbers input mode and bus names input mode.

7. Node-breaker Analysis

- New node-breaker specific APIs have been added to return the node number at which a terminal device (load, machine, fixed bus shunt, switched shunt, induction machine, OPF adjustable bus shunt) is connected. The corresponding new APIs are:

  - FETCH_LOAD_TERM_NODE
  - FETCH_MACHINE_TERM_NODE
  - FETCH_SHUNT_TERM_NODE
  - FETCH_SWS_SHUNT_TERM_NODE
  - FETCH_INDMACHINE_TERM_NODE
  - FETCH_ADJVAR_TERM_NODE

8. Short Circuit

Updated modeling of Non Conventional Source (NCS) fault current contributions in activities ASCC and SCMU.

- Updated how Time-Dependent Characteristics contribute to faults.
- Added modeling of Voltage-Dependent Characteristics contribution to faults.
- Sequence data record "Non Conventional Source Fault Contribution Data" has been re-organized in two sequence data records: "Non Conventional Source - Fault Contribution Characteristics Data" and "Non Conventional Source - Generator Data". Refer Data Format manual for details.
- Added section "Modeling Non Conventional Source Fault Contribution in Short Circuit Calculations" to Program Application Guide (PAG) in Chapter "Fault Analysis".

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- Added sequence data read change files "savnw_ncs_TI_rdch.seq" and "savnw_ncs_VI_rdch.seq" to Example folder. Updated Example folder "sample.seq" to add "Voltage-Dependent Characteristics" data records.

9. GIC

- Dialog "GIC >Download Efield grid data from Cloud..." updated to download "Fernberg 1D" and "Empirical 3D" Conductivity Model Geoelectric Field Map data json files from “Space Weather Prediction Center (SWPC), NOAA”.

10. Dynamic Simulation

- Added NERC GP3 model (Generic synchronous generator protection model).
- Added logic in the REGCA model to take care of applying the rate limit (rrpwr) correctly when the generator power is negative (e.g., when the generator is used to model a storage device which is in charging mode of operation).
- Added GovSteamEU model (Simplified Boiler and Steam Turbine with PID governor model - CIM model based on IEC 61970-302).
- Updated the CHVDC2U1 model to add internal integration logic.
- Updated the GENTPJ1 model to allow for the modeling of subtransient saliency (i.e., allowed for CON(J +10) to be different than CON(J+11)).
- Modified the PSLF to PSS®E dynamics data converter (WECCDS) for converting the following PSLF models: oel2c, uel2c, esdc4c, der_a, regc_b, reec_d.
- Added Machine reverse power relay model MCREPWU1.
- The baseload flag designation in the generator powerflow record is now honored.
- The following models were added for extracting phase and sequence voltages and currents:
  - GetBusPhsVLTU: This model can be attached to any bus, and can be used to retrieve the bus phase (A, B and C phase) voltage magnitude and angle.
  - GetBusSeqVLTU: This model can be attached to any bus, and can be used to retrieve the bus sequence (positive, negative and zero sequence) voltage magnitude and angle.
  - GetGenSeqCURU: This model can be attached to any generator, and can be used to retrieve the machine sequence (positive, negative and zero sequence) current magnitude and angle.
  - GetBrnSeqCURU: This model can be attached to any branch (including zero-impedance lines, 2-winding transformer and branches), and can be used to retrieve the branch sequence (positive, negative and zero sequence) current magnitude and angle. This model cannot be attached to a 3-winding transformer.

11. Optimal Power Flow

- Various OPF report tables have been updated to accommodate larger numerical values.

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- New OPF APIs have been added to provide the ability to change the identifier (ID) of an OPF adjustable bus shunt, and a branch/2-winding transformer or 3-winding transformer flow constraint. The corresponding new APIs are:
  - OPF_CHANGE_ADJVAR_ID
  - OPF_CHANGE_BRN_ID
  - OPF_CHANGE_3WT_ID

12. Slider Diagrams
    - Added the option of enabling/disabling the feature that snaps the cursor to the line when adding knee points to lines.
13. Processing Vendor Neutral Contingency and RAS definitions
    - Execution time performance has been enhanced. In various tests:
      - READ_RAS saw improvements of 53.4% to 98.4% from PSS@E-35.1.0 and PSS®E-35.2
      - VALIDATE_RAS saw improvements of 52.1% to 98.5% from PSS@E-35.1.0 and PSS®E-35.2
      - ACCC_RAS saw improvements of 24.1% to 99.5% from PSS@E-35.1.0 and PSS®E-35.2

      The amount of improvement depended greatly on the size and content of the case and the auxiliary file.
14. WECCLF Converter
    - None
15. Plot Package
    - Channel File filtering by channel identifier, number, quantity type and bus number has been added to the PlotTree, as well as a reset feature.
    - Added QuickPlot display for multiple filtered channels.
    - Voltage Violations QuickPlot display now has a zoom feature and an improved look and feel.
    - Added an option to the voltage violations dialog so colors used to indicate a violation, no violation etc. can be customized.
    - Users can now turn on/off chart legend, title and axes captions from a pop-up menu.
    - Charts on a page can now be resized.
    - Charts can be dragged around within a page.
    - The ability to customize the number of rows/columns and margins on a page has been added and this is recordable.
    - The ability to print an active page with multiple charts as well as record it has been introduced.

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- The user can now position and place charts on a page as desired for WYSIWYG printing and record this activity.

5.4. Aha! Ideas Portal

The following items from the Aha! Ideas Portal are included in this release:

1. PSS®E-I-310: SUB/MON/CON documentation into separate document, very hard to find under DFAX. The SUB/MON/CON documentation has now been moved from POM.pdf to CheatSheets.pdf.
2. PSS®E-I-446: Lock the Network Data Spreadsheet row height. A new setting has been added to Edit > Preferences - "General" tab - Spreadsheet interactions - Enable row height resizing. Toggling this will enable/disable spreadsheet row resizing. It is recommended that all spreadsheets be reset to defaults before toggling this setting for the first time.
3. PSS®E-I-863: Sortable, filterable substation selection menu.

5.5. Program Corrections

The following program corrections appear in this release:

1. Power Flow
   - Fixed an issue when the Percent of Voltage band for switched shunts (SWVBND) is less than 100, some switched shunts do not regulate their voltages even when the voltages are outside of control limits. The issue only exists between 34.2 and 35.1.
   - Corrected output messages and errors when reading switched shunt data. At times the wrong message was being output. This only affected prior releases of version 35.
   - Fixed a potential crash in GOUT. This resulted from joining (JOIN) two buses in which one bus had been recently added (such as through SPLT)and at least one of the buses to be joined had a 3-winding transformer connected.
   - Corrected an issue in the Saved Case file (.SAV) in which VSC or multi-term DC line converters connected to a bus within a substation had the potential for the converter's node terminal connection to not be saved in the case and thus not restored when the case is read back in. This typically only affected the last line read. It had the potential however of causing a crash when the node-breaker topology processor was run. Prior to this fix the remedy was to manually reconnect the VSC or multi-term DC converter to a node of the bus and save the case in Raw data file format.
2. OPF
   - The transformer tap value printed in the OPF Linear Constraint Report for transformer variables has been corrected. This fix addresses a potential crash.
3. PSCOPF
   - Fixed an issue when the system swing bus (type 3 bus) was connected to zero impedance lines, the results were not correct
   - Fixed an issue when the system swing bus (type 3 bus) was connected to phase shift transformers and the transformers were regulated by PSCOPF, the results were not correct.

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4. GIC

   - None

5. Dynamic Simulation

   - Bug fix in INTFLW T model (this bug would show as a crash during initialization).
   - Bug Fix dumping out the channel information using activity DMPC (this bug did not exist in PSS®E versions prior to 35.0, but got introduced at version 35.0 due to conversion to 64 bit architecture).
   - Bug fix in doing the DOCU and DYDA of “Bus Other” type model. As a result of this bug, DOCU and/or DYDA of “Bus Other” type models will output just one instance of this model type even if the case has multiple instances of the same.
   - Fixed a typo in the message output of VTGDCAT and VTGTPAT models (word IGNORED was spelled as IGNOREDD).
   - In the GENTPJ1 model, fixed an issue regarding the handling of implicit GSU (i.e., when RTRAN and/or XTRAN are greater than zero).
   - Corrected the VAR(L) description in the pitch controller model WTPTA1 documentation.
   - Fixed an issue in using python APIs for setting the options under Dynamics>Simulation Options.
   - Fixed an Issue in writing out the model name in the relay characteristic file ("rlc" file).
   - Fixed an issue in the calculation of the output of the lag-block of HVDCAT model.
   - Fixed an issue in the OEL2CU1 model data sheet - the CON description of CONs J+36 through J+39 were incorrect.
   - Fixed a bug in the API CCTMIND_MSCO. As a result of this bug, the model status was not being retrieved correctly.
   - In the turbine governor model TGOV1, the warning message regarding the mechanical power (PMECH) being initialized out of limits will be displayed only if PMECH is greater than the upper limit or less than lower limit. Earlier the message was being put out even if PMECH was initialized exactly at the upper or lower limit.
   - Fixed an issue in activity DYR when reading the character ICONs of CRANIT and VFT1 models.
   - Fixed an issue in the PSLF to PSS®E dynamics data converter (WECCDS) – in converting PSLF models esdc4b and esdc4c, if the derivative controller time constant (Tdr) is zero, or if the rate feedback time constant (Tf) is zero, then in converting these models to DC4B and DC4C, PSS®E will set the corresponding gains (Kdr, Kf) to zero and set Tdr and Tf to some number (say) 1.0. This is done to avoid division by zero in the PSS®E models.
   - WECCDS conversion of wt2g model has been modified to use default power-slip data if PSLF wt2e model is not found in the dyd file. This is to prevent “junk” data from being written out if the PSLF wt2e model is not found by the WECDS conversion program.
   - In the AC7B and AC7C models, the ‘initialized out of limits’ check has been changed to not put out the message if the variable initializes exactly at the limit.

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- The generator model consistency check has been updated to not put out an error if the generator model is GENTPJ1.
- Bug fix in CMLDxxDGU2 (DER_A+composite load model). This bug was due to wrong CON values were being used to check if the Motor D terminal voltage was less than Vstall(Adjusted). Additionally, there were issues with simulating the reclose of 3-phase Motors A, B and C.
- Minor changes in output messages from CMLDxxU2 and CMLDxxDGU2 models.
- Fixed a messaging issue in the CMLDxxDGU2 (xx could be BL/AR/OW/ZN/AL) model when the P/Q ratio is less than the specified PQratio threshold.
- Fixed an issue in the DC4B model related to the calculation and application of regulator (STATE(K+3)) lower limit which is the limit shown as VRMIN*VT in the DC4B model block diagram.
- Bug fix in the CHVDC2U1dyr syntax in the model documentation.
- Bug fix in Dynamics GUI when changing the status of owner based load and load relay models.
- Bug fix in the interactive (via GUI) addition of the IEEEG1D turbine governor model.
- Bug fix in turbine governor models with deadband (HYGOVD, PIDGOVD, TGOV1D, WPIDHYD, GASTD, GAST2AD, GASTWDD) – the input to the turbine factor should be speed deviation rather than the output of the speed deadband.
- Bug fix in DEGOV1D model; the output has to be multiplied by speed rather than the output after deadband.
- Fixed an issue when doing DOCU and DYDA output when the DOCU and DYDA outputs were being sent to report window.
- Bug fix in WT12T1, WTDTA1 and WT3T1 models in accounting for the parameter DAMP at initialization. As a result of this bug, these models could show initialization issues with STATE(K+2) when DAMP is greater than zero and if the initial generator speed is not 1 pu.

6. Node-breaker

- Correction made to plant totals when the status of a machine is changed through breaker actions.

7. Contingency Analysis

- Fixed an issue when a branch connected to a system swing bus (type 3 bus) is tripped by a contingency and the FDNS method is selected for contingency analysis, the contingency analysis may crash.
- Corrected an issue in which PSS®E would abort during the writing of the ACCC solution output file. This would most typically occur when executing multiple ACCC runs within the same PSS®E session.

8. Slider Diagrams

- Fixed an issue when attempting to open a recent file listed in the File menu was freezing PSS®E, if the file was not found.
- Fixed a crash issue when drag-and-dropping an SLD from one study group to another.

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- Corrected an issue in which PSS®E was always asking if the user wants to save the case when another case is opened or PSS®E is closed, even if the case has not been modified.
- Fixed an issue with diagram titles that could not be removed once set in PSS®E-35.
- Fixed an issue with the summation labels, labels were not being updated automatically.
- Fixed an issue with annotation labels that were becoming visible only at certain zoom levels in a diagram.
- Fixed issues with exporting/importing Program settings/preferences.
- Fixed issues with PSS®E-33 cases which were not automatically set to auto-load when added to study groups and could not be set to auto-load manually in the Study Explorer.
- Fixed an issue when using Join Bus from G-OUT new busbars were being created and, connections were confusing.
- Fixed an issue After changing the symbol of a switched shunt in a diagram, you had to manually refresh the diagram for the change to show.
- Corrected an issue in which ID's of switched shunts were not shown in any diagrams even if all other equipment ID's were shown.
- Fixed an issue When selecting a whole area of a diagram the "selection square" was not disappearing by itself.
- Fixed an issue when resizing busbars on one end only was not working properly.
- Corrected an issue in which labels were becoming unselectable in certain areas of a diagram.
- Fixed an issue when trying to "Ungroup" diagram objects was leading to PSS®E crashing.
- Corrected an issue in which link connection positions were not changing from one end.
- Corrected an issue in which default diagram properties could not be saved.
- Corrected an issue in which visible value was always staying checked in item properties.
- Fixed an issue in which after the power flow analysis has been made, the diagram was not being updated by itself.
- Fixed an issue in which after the power flow analysis has been made, the diagram was updating by itself.
- Fixed an issue in which sometimes the diagram queries to the engine would result in an "Invalid sub-station number" error, causing the error to come up repeatedly for every item on the diagram.
- Fixed an issue when tab line functionality has been run, the application was creating an unbound link.
- Fixed an issue in which the values on an unbound branch were not showing up after bound it.
- Fixed an issue in which network tree was not being updated after status changes.

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- Fixed an issue when converting diagram files created by PSS@E 33, there was a mismatch between the diagram and the "Multisection line reporting" menu item.
- Some issues with loading diagram files created by PSS@E 33 are solved. PSS@E 35 was crashing while opening these files.

9. WECCLF Converter
   - None
10. Python
    - Improved display and data handling for Unicode and non-ASCII (also called extended ASCII) characters.
11. Bug fix in writing PSS®E version 29 RAW and SEQ files for multiple switched shunts on a bus.
12. Plot Package
    - Two channel files with the same name cannot be open simultaneously.
    - Fixed a bug in Voltage Violations calculations and save/restore CSV files.
    - Added a check for invalid channel values in file before plotting.
    - Fixed a bug when closing channel files.
    - Fixed a bug with the zoom option on the chart's pop-up menu.
    - Fixed a crash when the user attempts to drag/drop multiple channels simultaneously.

5.6. Known Issues

The following issues have been identified and known to exist in this release:

- All recent Intel Graphics drivers released since May 15, 2019 AND before August 12, 2020 appear to have issues which result in the incorrect display of Slider diagrams. Do not update your Intel Graphics driver to anything above 26.20.100.6709 OR below 23.20.16.7584. If you have a version that falls in this gap, please uninstall that version and install version 23.20.16.7584 or greater.

  Alternatively, if you have a discrete graphics card from AMD or NVIDIA, make sure to configure your machine so that the discrete card is used for PSS®E instead of the integrated Intel processor. If you do not wish to update your drivers, you can also choose to turn off Hardware Acceleration by going to [Edit->Preferences...], [Diagram] tab, [Advanced Global Settings] tab. This will force the diagrams to use the Microsoft provided software-only rendering, but this is much slower than using a graphics processor, and much slower in Windows 10.

- When using a CodeMeter network license, the computer acting as the license manager will need the CodeMeter Runtime upgraded to a minimum of 6.80. Download and install the latest version of the CodeMeter User Runtime from www.CodeMeter.com.

- Use of ACCC_PARALLEL, ACCC_PARALLEL_2, ACCC_WITH_TRIP_PARALLEL, N11_ACCC_PARALLEL, MACCC_PARALLEL, or MACCC_WITH_TRIP_PARALLEL are not permitted with a case containing substation data. This is being addressed and will be resolved in a future release.

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- Currently, it is not possible to Filter Report Tables. Attempting to filter Report Tables will result in a crash.
- Currently, it is not possible to query existing Report Tables using Python. This functionality will be added in a future release.
- Currently, only select activities use the Report Table format. The Report Table format will be rolled out to include additional activities in future releases.
- Not all Advanced Linear Analysis/MUST reports are currently written to Report Tables. Missing reports will be added in future releases.
- There are several known issues when switching back and forth repeatedly between Advanced Linear Analysis/MUST and PSS®E functionality in the same session. These issues are being addressed and will be fixed in future releases.
- Text containing non-ASCII characters may produce an error in Python or may be represented incorrectly with Python data retrieval APIs. This issue will be resolved in a future release.
- Occasionally, the value of DynOutx in the HKEY_CURRENT_USER "Dynamic Engine Options" registry "hive" will become corrupted. When this occurs you will observe RWFIL errors when starting dynamic simulations. This error can be cleared by deleting the DynOutx Key from the registry and letting the program reinitialize this value upon startup.

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Chapter 6
PSS®E-35.1.0 Release Notes

May 2020

6.1. Discontinued Modules

The Measurement Interface and the Data Visualization and Reporting Module (DVRM), have both been dropped from this release, PSS®E-35.1. If you are currently licensed for these modules you may continue to use these modules in your PSS®E-35.0 installation.

If you wish to continue using your Measurement Interface and DVRM modules, do *NOT* un-install PSS®E-35.0 before installing PSS®E-35.1. Failure to preserve your PSS®E-35.0 installation will result in the loss of these modules.

6.2. Program Enhancements

The following program enhancements appear in this release:

1. The following Report Tables have been added to PSS®E:
    a. Power flow (POUT) report
    b. Power flow wide format (LOUT) report
    c. Power flow wide format with Amps (LAMP) report
    d. Load Reduction report
    e. Distribution factors (OTDF) report
    f. Transmission Interchange Limits (TLTG) report
    g. Area report
    h. Zone report
    i. Area in Zone report
    j. Zone in Area report
2. MUST

    The following Report Tables have been added to the MUST module along with the respective APIs to generate the reports programatically:

    a. FCITC Single Study report

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b. Base Case Violations report

c. Subsystem Adjustment Details report

d. FCITC Multiple Transfer report

e. AC FCITC Single Study report

f. AC FCITC Multiple Transfer report

g. Subsystem Sensitivities report

h. Subsystem Impact Sensitivities report

i. All Subsystem Sensitivities report

j. Transaction Sensitivities report

k. All Transaction Sensitivities report

- Area/Zone "Bubble" diagrams are now available in the MUST module.
- The Exclude feature in the MUST Module has been reworked to allow a more interactive approach to excluding monitored elements. Elements may now be excluded singularly or in groups, either from a dialog or through specific APIs as needed.
- A text based filtering capability has been added for all selection dialogs in MUST (e.g. select monitored elements to exclude).
- APIs have been added to MUST that allow selecting monitored elements by cross-referencing for more convenient retrieval. (e.g. return a contingency number by providing the contingency name and vice-versa).
- The MUST Excel Add-in is now available for importing MUST reports generated in the classic format. This Add-in also includes a feature for comparing Contingency Analysis Reports for imported classic MUST Reports.

3. RAWX and Extension Data

- Added support for custom user tables in RAWX files.
- Partial datasets can be imported from RAWX files by setting the new case flag (ic field in caseid parameters) to 1
- RAWX tables can can be imported into PSS®E from Comma-separated Values (CSV) files (see API CSV_TO_RAWX). All built-in and custom fields from the data table are exported.
- RAWX tables can be exported from PSS®E to Comma-separated Values (CSV) files (see API RAWX_TO_CSV). These imported tables are treated as partial datasets, similar to the Read and Change (RDCH) activity, and they may include custom fields.
- Added APIs for reading and writing all extension data, including custom tables (see APIs with prefixes GET_EXT_ and SET_EXT_).

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- Improvements to READX and RAWDX performance, error handling, and reporting.
- Added support for custom ordering of parameter sets and tables in RAWX file.

4. Contingency Analysis

- The API rxpath is provided to find the shortest path from the starting bus to the ending bus in terms of number of buses or the total impedance. The API can be called in python.
- Contingency analysis solution parameters are added and used by contingency analysis activities (AC contingency analysis, Multiple level contingency analysis, N-1-1 and PSCOPF). With the solution options of the activities, they initiate contingency analysis, control the procedure to evaluate a contingency and select results displayed in the spreadsheet table and reports. The parameters include contingency analysis rating sets, AC contingency screening options and contingency analysis solution options.
- Robust Power Flow Solution is integrated with AC contingency analysis and N-1-1 contingency analysis. If a contingency fails to converge with a pre-defined method and options, the contingency is resolved by the robust power flow solution if the Robust Power Flow solution option is enabled.
- AC contingency screen option is added and used to select from the given list the contingencies that need a full AC power flow solution. The screen method is based on the DC power flow network.
- Base loaded machines are supported in AC contingency analysis. Users can select one of three options to handle base loaded machines dispatched by the generation dispatch activity and the corrective action activity in a contingency analysis.
- Off-line generator control is available in DC corrective action.
- Robust Power Flow Solution is integrated with AC corrective action activity, where the control adjustments are verified by Robust Power flow solution if the regular Full-Newton Raphson solution fails to converge.
- The total MW generation dispatch limit can be added in corrective action activity. The total generation that is dispatched either up or down has to be less than the limit.

5. Short Circuit

- The NCSFC machine is considered as open circuit in the zero sequence network when its Z0 is zero. The zero sequence impedance Z0 (=ZR0+jZX0DV) is provided on machine sequence data record.
- DataFormat Manual update-For those machines that do not include the step-up transformer as part of the generator data (i.e., XTRAN is zero), a zero sequence impedance (ZR0, ZR0DV) of zero and grounding impedance (ZRG, ZXG) of zero results in the machine being treated as an open circuit in the zero sequence.

6. GIC

- Added "efieldgrid" event to use non-uniform near real-time geoelectric field data to calculate branch induced voltages due to GMD. The efield grid map data is available from USGS/SWPC web portal in JSON file format or can be provided as CSV or JSON format file.

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- Added activity to download the latest or any previous time efield grid map data from USGS/SWPC web portal.
- Added activity to read .json and .csv format efield grid map data files.

- For Benchmark+Supplemental Event calculation when Supplemental Subsystem is defined by bus subsystem, branch was considered in supplemental event Efield when both buses were in supplemental event subsystem. This is changed to:
  - when option brn_seg_efld=0, branch is considered in supplemental event Efield when both buses are in supplemental event subsystem.
  - when option brn_seg_efld=1, branch is considered in supplemental event Efield when either of buses is in supplemental event subsystem.
- The GIC(t) .csv file will have only applicable event waveform(s). It will have both benchmark and supplemental event waveforms only when both events are simulated.
- The GIC(t) .csv file now shows the event applicable to various transformers as shown below: [benchmark], [supplemental]
  - TransformerID,TransformerName,GIC_E(A/ph),GIC_N(A/ph),GIC_t(A/ph)
  - 152_153_T3 [benchmark],MID LTC, 0.906560, 2.951407, ...
  - 152_3021_T4 [benchmark],WDUM DC, 2.167822, 5.948224, ...
  - 154_9154_W1 [supplemental],WTG1XMER, 0.098521, 0.199759, ...
- Updated GICMAPS module: Added following two methods to gicmapsobj.

  - report_supp_box_line_segments(**kwds)

  Create text report of study subsystem branch segments inside and outside of supplemental event moving box.

  - report_network_element_events(**kwds)

  Create text report of showing type of event applied to substations, buses and transformers. [With this you should be able to see if the event applied in GIC(t) .csv file is correct.]
- Added function to show usage of GICMAPS, how to draw moving box, create text reports to gic_demo.py file in Example folder.

7. Dynamic Simulation

- WECCDS (PSLF dyd to PSS®E dyr) converter enhanced to convert the following PSLF models: lhfrt, lhvrt, h6e, reec_c.
- Added dynamic load model CMLDxxDGU2 model (where xx could be BL, AR, OW, ZN or AL). This model has the DER_A added as part of the composite load model. The DER_A included int this model uses the MW and MVAR specified in the load record of PSS®E powerflow.

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- IEEE 421.5 Overexcitation Limiter model OEL5C (called OEL5CU1 in PSS®E).

8. Slider Diagrams

   - Slider diagram rendering performance improved by the modernization of infrastructure. Achieved smoother zoom in/out and panning behavior.

9. WECCLF Converter

   - Added "Robust Solution" method. Default solution method set to RSOL.
   - Allow BMAG in PSLF/transformer data to be negative (even though PSS®E produces warning messages for negative MAG2).
   - PSLF Breaker data to PSS®E System Switching Device Data.
   - Set bus Maximum and Minimum emergency voltage limits to default values when PSLF values are 0.0 or equal.
   - Make branch, system switching device and transformer names to be unique.
   - Set transformer max and min voltage limits to defaults to avoid progress message warnings.
   - Set transformer rmax and rmin to defaults to avoid progress message warnings.
   - In PSLF Transformer terminated lines (PSLF type=1), values of PSLF data items gi, tf and tt checked and if all are zeros, line is treated as normal PSS®E line.
   - PSS®E Facts device name assigned and made unique when not provided in PSLF data.
   - PSS®E Switched Shunt vslo and vshi limits set as below.

     For PSLF SVD type 3 and 4, when vband<=0, vrefmin and vrefmax sets voltage control range.

     For PSLF SVD type 5 and 7, vrefmin and vrefmax sets voltage control range.

   - For transformer phase shifter control mode PSLF ctyp=[4,14,-4,-14] and PSS®E COD=[3,-3] set PSS®E [VMA, VMI] to PSLF [vtmax, vtmin] as is without checking any limits.

6.3. Program Corrections

The following program corrections appear in this release:

1. OPF

   - Added a check to ensure that OPF data has been initialized prior to OPF API commands being run.
   - Cleaned up the OPF Interface Flow Report:
     - Ensuring that extraneous characters do not show up on participating branches that do not have any flow
     - Added a header and constraint information for each individual interface flow constraint

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- Fixed an issue in the OPF Adjustable Load table to ensure that all impacted loads are printed out.
- The OPF Linear Constraint data now allows an external table ID number of zero (0).
- For OPF solutions involving the minimization of adjustable bus shunts, a correction has been made to the calculated objective cost value displayed at the end of the OPF solution iteration log. An incorrect value could occur when var records had a maximum var limit that was positive and a minimum limit that was negative. This only affected the "Minimum shunt var objective" value that was displayed, not the individual var values that were calculated as a result of the OPF solution and displayed in the Added Shunts (MVar) table in the OPF report.
- For OPF branch flow constraints on 3 winding transformers, the data spreadsheet now correctly displays the 3 unique buses, with the winding of interest listed first.
- The OPF Report has been cleaned up to ensure that form feed characters do not erroneously appear.
- Fixed an issue when saving an OPF Raw File in which the participating branch IDs of interface flow constraints may be incorrect.

2. PSCOPF

- Fixed an issue with PSCOPF where a N-1-1 contingency may be skipped if the two N-1s come from two different contingency description data files.

3. GIC

- When tielvls>0, due to bug in how tielevel was processed, some three winding transformer buses could have been missed from GIC study subsystem.
- When boundary_trn>0, due to bug in processing boundary transformer buses, some boundary transformers buses were not added to GIC study subsystem. Hence some of the two and three winding transformers could have been missed from GIC study subsystem (transformer GICs flow not shown in the report).
- When both benchmark and supplemental events were simulated, the transformer GIC(t) response was calculated considering only benchmark event. This bug is fixed to consider the transformer location and supplemental event criterion [subsystem or moving box] to determine the event applicable to the transformer. The GIC(t) response is then calculated considering that event.

4. Dynamic Simulation

- Fixed an issue in activity DYR in reading in the character ICON in the PSS®E standard library model PLBVF1.
- Fixed an issue in the DOCU check logic of WT3G2 model when the number of lumped wind turbines (specified in ICON(M)) times Prate (specified in CON(J+5)) is greater than the MVA base of the machine.
- Fixed an issue in writing out dyr record (activity DYDA) of SAT2T and PLBVF1 models.
- Fixed CON(J+16) description of REPCA1 model - should have been integral gain (was shown as proportional gain).

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- Fixed a typo in the CON description of CON(J+7) of the REGCA1 model.
- Fixed an issue related to reading in either 35.0 created dynamic snapshot or opening version 33 and 34 dynamic snapshot file in version 35 when the snapshot file contained any of the following CCT Type models: branch device models, 2-winding and 3-winding transformer device models, machine protection models, and 2-terminal dc line other model.
- Fixed an issue with block diagram of AC3C model (the PID controller and the two states associated with this PID block) were missing from the block diagram.
- Fixed an issue with DYDA output of DC4B and DC4C model (model was writing out one more line than necessary).
- Fixed an issue reading in dynamics dyr file containing switched shunt models.
- Fixed an issue in using switched shunt model APIs like CHANGE_SWSMOD_CON etc . This bug will show up when the API is being invoked with old model name but the model (while reading in through a dyr file) has been converted by PSS®E to the corresponding new model name.
- Fixed an issue with SGT8HM1 model. Some limits were not being calculated correctly and as a result the mechanical power output could become negative following a load shedding.
- Fixed an issue involving crash when opening model edit dialog for SVSMO2T3 model.
- Fixed an issue involving deletion of switched shunt models following a switching operation of mechanically switched shunt.
- Following modification in CMLDXXU2 (xx could be BL, OW, ZN, AR, AL) model: allowed the ability to set LTC flag (CON(J+8)) to -1 as well. With this change, the interpretation of the LTC flag specified in CON(J+8) is as follows. If this CON is any value less than or equal to 0, the LTC logic will be implemented only during initialization but not during simulation. If the LTC flag is set to 1, then the LTC logic will be implemented during initialization and also during simulation.
- Fixed an issue in opening the 'Model tree view' of switched shunt models for those switched shunts that are OFF in powerflow.
- Fixed an issue in doing DOCU of switched shunt models.
- Fixed an issue involving crash when opening 'Models Tab' for viewing 'Renewable Auxiliary Control' and 'Renewable Torque Control' models.
- In the SWCAPT model, fixed an issue in writing out the Model Instance (MINS) value (this issue was observed as a crash in some simulations).
- In the BUDCZT model documentation, interchanged CONs J+5 and J+6 to keep it consistent with the logic as coded in the model.
- In the model AC7C, VARs L+2 and L+3 were not being assigned a value. This change would not affect the simulation results in any way.
- Fixed an issue in the REECC and REECB models, Kqv injection (Iqinj) should be non-zero irrespective of whether voltage_dip is true or not.

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- Fixed an issue related to the display of generator dynamic models in the dynamics data spreadsheet view for "infeed" machines (machines marked as "infeed" type in the generator powerflow data record).

- Fixed an issue in the AC7B and AC7C models - the limit check of the PID controller was being done after the HV and LV gates in position c. This PID controller output should be applied before the signals pass the position c HV and LV gates. This change would affect the simulation results only if the AC7B or AC7C models had UEL or OEL models applied at position c.

- Fixed an issue in the DYDA format statement of PVEU1 model. The format for writing out the first ICON of the model (which was the remote bus number) was not adequate as a result of which when attempting to write the bus numbers that are over 3 digits long, the output (dyr file) would contain **** instead of the actual bus number.

- When performing STRT (dynamics initialization), if users select the option to gnet generators for which no dynamic models have been provided, if PSS®E finds a generator for which there is no dynamic model attached and if the machine P and Q output both are zero, PSS®E will set these machines to out-of-service.

- Fixed an issue in the ICON(M+6) description in the PLNTBU1 model documentation. The ICON(M+6) should be set such that 1 stands for “enable”, 0 for “disable”.

- Fixed an issue in the ICON(M+6) description in the REPCA1 and REPCTA1 model documentation. The ICON(M+6) should be set such that 1 stands for “enable”, 0 for “disable”.

- Fixed an issue related to the display of generator dynamic models.

- Fixed an issue in activity DYR in reading in the character ICON in the PSS®E standard library model PLBVF1.

- Modified COMON4.INS file to remove FORTRAN continuation character. This will allow the COMON4.INS file to be used in dynamic models written in f90 syntax as well.

5. Node-breaker

- When reading in a Powerflow Raw Data File (.raw) containing substation data, spurious warning messages are no longer output. Additionally, when data error messages are encountered when reading the main substation data record, the remaining data records for that substation are skipped and the substation is not added.

- The status of terminal equipment (shunts, machines, loads) now reflects the status of the node to which it is connected. This addresses the issue where a node was taken out of service but the associated bus, or bus section, to which it belonged to was in service and hence the equipment connected to the out-of-service node also was treated as in service when it should have been out of service. In future releases the original values of the terminal connected equipment (shunts, loads, machines) will be preserved such that when their connected node is reactivated, the status of the connected equipment will return to its original status.

6. Contingency Analysis

- Corrected an issue with flows on windings of 3-winding transformers connected to nodes within a substation. The windings were erroneously appearing to not appearing in overload reports.

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- Corrected an issue in which voltage violations were erroneously being reported or not reported. This occurred in situations where a node within a substation becomes isolated due to switching device actions and is placed out of service but the connected device (load, machine, shunt) remained in service.

7. Slider Diagrams

   - Fixed an issue involving crash when switching a line that has knee points.
   - Fixed an issue on updating diagrams. After running an analysis, the diagram was not being refreshed unless the user clicks on the diagram.

8. WECCLF Converter

   - Set correct bus type (=1) for PSS®E system switching device buses converted from PSLF breaker branches. They were set as type=4 (disconnected) even when SWD branch was in-service. [Otherwise some breaker buses were isolated and power flow will not run (topology error).]
   - PSLF SVD to PSS®E Switched Shunt mappings changed as below.

     SVD ty=7 changed to PSS®E mode=1 (discrete).

     When SVD vs=0.0, set PSS®E vs=1.0.

     Updated PSS®E Switched Shunt limits to consider following.

     - When PSLF vband <= 0, the vrefmin and vrefmax values are used as the voltage regulating range.
     - SVD vrefmin and vrefmax are applicable to svd ty=3,4 when vband<=0.0.
     - SVD vrefmin and vrefmax are applicable to svd ty=5,6,7 when slow B is enabled.

9. Python

   - Corrected an issue that prevented certain Python scripts and wxPython scripts from launching when Python was not installed with the "Install for all users" option selected.
   - Exporting results to Excel will no longer display "python interpreter not found" when Python was not installed with the "Install for all users" option selected.
   - Opening Environment Manager from within PSS®E will no longer display "python interpreter not found" when Python was not installed with the "Install for all users" option selected.

6.4. Known Issues

The following issues have been identified and known to exist in this release:

- All recent Intel Graphics drivers released since 5/15/2019 AND before 18/12/20 appear to have issues which result in the incorrect display of slider diagrams. Do not update your Intel Graphics driver to anything above 26.20.100.6709 OR below 23.20.16.7584. If you have a version that falls in this gap, please uninstall that version and install version 23.20.16.7584 or greater. Alternatively, if you have a discrete graphics card from AMD or Nvidia, make sure to configure your machine so that the discrete card is used for PSS®E instead of the integrated Intel processor. If you do not wish to update your drivers, you can also choose to turn

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off Hardware Acceleration by going to [Edit->Preferences...], [Diagram] tab, [Advanced Global Settings] tab. This will force the diagrams to use the Microsoft provided software-only rendering, but this is much slower than using a graphics processor, and much slower in Windows 10.

- When using a CodeMeter network license, the computer acting as the license manager will need the CodeMeter runtime upgraded to a minimum of 6.80. Download and install the latest version of the CodeMeter User Runtime from www.CodeMeter.com.
- Use of ACCC_PARALLEL, ACCC_PARALLEL_2, ACCC_WITH_TRIP_PARALLEL, N11_ACCC_PARALLEL, MACCC_PARALLEL, or MACCC_WITH_TRIP_PARALLEL are not permitted with a case containing substation data. This is being addressed and will be resolved in a future release.
- Currently it is not possible to query existing Report Tables using Python. This functionality will be added in a future release.
- Currently, only select activities use the Report Table format. The Report Table format will be rolled out to include additional activities in future releases.
- Not all MUST as a Module reports are currently written to Report Tables. Missing reports will be added in future releases.
- There are several known issues when switching back and forth repeatedly between MUST and PSS®E functionality in the same session. These issues are being addressed and will be fixed in future releases.
- Text containing non-ascii characters may produce an error in Python or may be represented incorrectly with Python data retrieval APIs. This issue will be resolved in a future release.
- Occasionally, the value of DynOutx in the HKEY_CURRENT_USER "Dynamic Engine Options" registry "hive" will become corrupted. When this occurs you will observe RWFIL errors when starting dynamic simulations. This error can be cleared by deleting the DynOutx Key from the registry and letting the program reinitialize this value upon startup.

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Chapter 7
PSS®E-35.0.0 Release Notes

September 2019

7.1. Installation Notes

If you have installed *any* PSS®E-35 pre-releases, please make sure your User Example directory is up-to-date by selecting File > Reset Example Directory when first launching PSS®E.

7.2. Program Licensing

Please note that PSS®E now exclusively uses Wibu CodeMeter for program licensing. Sentinel dongles and activation strings are no longer supported.

7.3. Program Enhancements

The following program enhancements appear in this release:

1. PSS®E is now a 64 bit application. This means that it can now handle larger contingency cases and users will experience performance improvements in some operations. MUST as a Module will take advantage of the increased available memory to allow larger contingency cases to be analyzed as well.
2. MUST is now integrated into the PSS®E application directly allowing you to perform both types of analysis from a single application.
3. PSS®E now supports multiple switched shunts at a bus. These switched shunts are identified with a bus number and a 2 character ID. When pre PSS®E-35 cases with existing switched shunts are read in, they are automatically assigned an ID of "1".
4. Each block of a switched shunt now has its own status flag for moving blocks in/out of service.
5. Support for Python 3.7 has been added and is now the default Python version. If Python 2.7 is desired add '-pyver 27' to the PSS®E 35 desktop icon Properties>Target command or to the psse line mode command.
    a. Right click the shortcut
    b. Select properties
    c. Select the "Shortcut" tab
    d. At the end of the text in the "Target" field, add -pyver 27
    e. Click "Apply" and "OK" to apply the changes
6. An extensible RAW file is now supported, RAWX. User created data can now be added to the working case and queried. When writing the network data to a RAWX file, all user created data is retained and output.

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7. A new reporting format, the Report Table, has been introduced to Contingency analysis and Probabilistic Reliability analysis. This format allows the reports to be presented as both tabular text reports and spreadsheet reports. In addition, the report tables can generate CSV files directly and be exported to Excel for additional post processing.

8. New AC Contingency analysis reports of the Contingency Summary, Overload Violations and Voltage Violations are automatically produced. In Multi-level AC Contingency analysis with Corrective Actions, an additional Adjustment report is produced.

9. Machine capability curve data is now preserved in the SAV and RAW files allowing it to be maintained along with the network data in a single file. The power flow solutions update the reactive power limits, as determined from its capability curve, of any machine for which a capability curve has been specified. Full reporting functionality exists for the curve data.

10. The number of available Owners has been increased to 9,999.

11. For node-breaker actions, an "original status" value is automatically preserved for all branch devices. This original status is initialized to the status of the device when the network case is read in and any subsequent explicit changes to the status that are not related to node-breaker actions will update or set the original status to that of present status.

    If, during node-breaker switching actions, a branch device is taken out of service or blocked, then the original status will be preserved while the device status will be set to out-of-service or blocked. Likewise, if during node-breaker switching actions, a branch device that is currently set as out-of-service or blocked can be placed back in service, then the value of the original status will be used to set the device status. If the status was originally out-of-service or blocked, then it will remain as such.

12. A new original status column is displayed in List Data reports for branches, line shunts, two and three winding transformers, FACTS devices and DC lines. If the original status differs from the present status, then the value of the original status will appear next to the current status (ST) or mode (MODE, MDC) column, under the column heading "OR". If there is no difference between the original status and the present status, then the original status column field for that device will be blank.

13. A baseload flag item has been added to the machine record. Although not currently supported in the simulation engines, future releases of PSS®E-35 will allow users to specify generators that cannot be redispatched during AC contingency analysis.

14. GIC Module

    - GIC is now is compatible with node-breaker elements in a working case.

    - GIC data is now saved in the network Saved Case file (.sav).

    - New GIC data file version=5 is added which uses substation location and connection/configuration information from the working case.

    - Added APIs to read and write GIC data in the format of GIC data file versions 4 and 5. Need to read GIC data before GIC activity is run.

    - GIC data file version 4, which contains substation location and bus substation data, can only be read into a working case that does not contain any Node-Breaker configuration data. Reading this GIC data adds a "Single Bus" substation configuration for each bus-substation record to the working case. De-

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pending on the size of the network and number of substations added, this process could take considerable CPU time. Save the updated working case to a .sav file and GIC data to a GIC data file version 5 (.gic). Then use this .sav and .gic files in any GIC analysis.

- Added GIC data spreadsheet views from PSS®E GUI.
- Added number of APIs to edit/change GIC data from GIC data spreadsheets.

15. Short Circuit Module

- As previously described, an original status column "OR", related to node-breaker switching actions, has been added to Short Circuit List reports for branches, line shunts and two and three winding transformers.

16. New Dynamic Model Additions and Dynamic Engine Enhancements

- Added AC6C2 model. This is a revised version of AC6C to allow users to specify voltage dependency of limit on regulator output.
- Added WT12A1U_B model. This is the model referred to by NERC and WECC as the revised pitch control model wt1p.
- Changes in OSSCAN functionality. The Out-of-step scan functionality (under Simulation Options) now has an option to exclude transformers.
- Added IEEE 421.5 2016 OEL2C model.
- Added Siemens SGT-2000E Gast Turbine model.
- Added Siemens SGT-8000H single and multi-shaft gas turbine model.
- Maximum number of wind machines increased to 20,000 at 150,000 bus sizing.
- Maximum number of CCT models increased to 150,000 for all bus sizing 50,000 and above.
- Increased the maximum number of line relay model connections (MXRLCX) by a factor of 10. This means, at the maximum bus sizing of 200,000 buses, the value of MXRLCX, which was earlier set at 4000, has now been increased to 40,000.
- The following switched shunt models - ABBSVC2, CHSVCT2, CSSCST2, SWSHNT2, were modified to take into account the ability to model status of individual switched shunt blocks.
- Three new threshold values for use in load models: Load MW threshold, Load P by Q (absolute value of P over Q) threshold, and Load bus voltage threshold. These threshold values can be used in dynamic load models to decide whether the load model should or should not be applied. The threshold values are accessible under Dynamics>Simulation Options, and can be accessed and/or set via automation (idv and python) scripts.
- A new threshold value for use in frequency relay model: minimum pu voltage for frequency measurement. This threshold value is accessible under Dynamics>Simulation Options, and can be accessed and/or set via automation (idv and python) scripts.
- WECCDS (PSLF dyd to PSS®E dyr) converter enhanced to convert PSLF models lhfrt and lhvrt.

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- The following models which were supplied as PSS®E Library user models (i.e., models required keyword 'USRMDL' in the dyr records) have been converted to PSS®E Library Standard models with new model names. The table below shows the old model name and the corresponding version 35 model names.

Pre-version PSS®E-35 Model Name	PSS®E-35 Model Name
PLBVFU1	PLBVF1
PSS2CU1	PSS2C
PSS6CU1	PSS6C
PSS7CU1	PSS7C
UEL2CU1	UEL2C
OEL2CU1	OEL2C
AC11CU1	AC11C
AC6CU2	AC6C2
AC7CU1	AC7C
AC8CU1	AC8C
AC9CU1	AC9C
DC4CU1	DC4C
ST6CU1	ST6C
ST8CU1	ST8C
ST9CU1	ST9C
ST10C1	ST10C
ST4CU1	ST4C
TGOV1DU	TGOV1D
GASTDU	GASTD
HYGOVDU	HYGOVD
IEESGODU	IEESGOD
IEEEG3DU	IEEEG3D
TGOV3DU	TGOV3D
WPIDHYDU	WPIDHYD
GAST2ADU	GAST2AD
GASTWDDU	GASTWDD
WESGOVDU	WESGOVD
DEGOV1DU	DEGOV1D
PIDGOV1DU	PIDGOV1D
GGOV1DU	GGOV1D
H6EU1	H6E1
HYG3U1	HYG3
SGT2EU1	SGT2E1
SGT8HMU1	SGT8HM1
SITGTU1	SITGT1
IEEEG1DU1	IEEEG1D

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Pre-version PSS®E-35 Model Name	PSS®E-35 Model Name
HYGOV2DU	HYGOV2D
DERAU1	DERA1
REECCU1	REEC1
CCOMP4U1	CCOMP4A

Although the model names have been changed, users can continue using the old model names as is (i.e., users do not have to change the dyr data file). PSS®E will take care of calling the new models internally. Should the user decide to write out the dyr records of any of these models (via activity DYDA), PSS®E will output the dyr records with the revised model names.

17. The maximum number of substations, nodes and switching devices have been increased. The maximum number of substations is set to half the maximum number of buses and the maximum number of nodes and the maximum number of switching devices are each set to 10 times the maximum number of substations.

	1,000 Buses	4,000 Buses	12,000 Buses	50,000 Buses	200,000 Buses
Substations	500	2,000	6,000	25,000	100,000
Nodes	5,000	20,000	60,000	250,000	1,000,000
Switching devices	5,000	20,000	60,000	250,000	1,000,000

18. When comparing powerflow cases (DIFF), there are new options to check for differences in bus substation assignments, substation node statuses, substation switching device statuses and substation switching device loadings.

19. Intra-substation loop flows can be displayed on substation flow reports, through API SPOUT_2, and can be highlighted on substation diagrams.

20. When saving in PSS®E Raw format, there is a new option for producing an RDCH file that excludes all substation data records.

21. When saving in PSS®E Raw format, there is a new option for producing an RDCH file that contains only substation data. The substations saved within the RDCH file can be further filtered using the bus subsystem selector.

22. Added Switching Device violations to the AcccBrwsGrid application.

23. The Environment Manager, used for compiling and linking user models has been updated to 9.0.0 to support building models for PSS®E-35

24. Updated the MicroSoft Visual Studio project, dsusr.sln, used to build a debuggable model for development and testing.

25. Updated the PSLF Power Flow and Dynamic converters to support PSLF Version 18, 19 and 21 files.

26. Updated RAW and SEQ converters to support output of PSS®E-34 and previous version files.

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7.4. Changes in Dynamics when Modeling Multiple Switched Shunts

At version 35, there are changes in the dynamic simulation engine, models and APIs to account for the presence of multiple switched at the same bus. As a result of this change, although all existing switched shunt model related APIs (i.e., pre-version 35 dynamic simulation switched shunt related APIs in the users idv and/or python scripts) would work as is, it is to be noted that the old APIs will always get applied to the first switched shunt at that bus (based on the order in which the switched shunt data is specified in the powerflow raw data file).

Because of the above changes, dyr files containing dynamics data for Switched Shunt models and switched shunt "other" type models may need have to be modified. Additionally, dynamic Switched Shunt models and switched shunt "other" type models will have to be modified. Pre-version 35 dynamic snapshots will be read by PSS®E and the switched shunt models will be applied to the first switched shunt at the bus at which the switched shunt is connected.

Changes to the dyr records of switched shunt and switched shunt other models are outlined below.

- If the dyr record has the keyword 'USRSWS', these can be used as is. However, the model will get applied to the first switched shunt at that bus.
- If the dyr record uses the new keyword 'USRSWS2', there will be a provision to specify the switched shunt id. For details of 'USRSWS2', refer to Program Operation Manual (and search for USRSWS2).
- Irrespective of whether the dyr file uses the old keyword 'USRSWS' or the new keyword 'USRSWS2' to designate the model as being attached to a switched shunt, changes will have to be made in the PSS®E user-written dynamic models of switched shunt and switched shunt "other" type models. The changes to made are as outlined below.

    1. The first argument of the model subroutine which used to be the bus index (say IB) will have to be changed to switched shunt index (say ISW). Note that ISW is to be declared as a FORTRAN integer variable.

    2. Since the switched shunt index (ISW) is an input into the model, any references to NSWICH (which was used to obtain the switched shunt index earlier) need to be removed.

    3. Knowing ISW, the bus index IB at which the switched shunt is connected can be obtained as NUMSWS(ISW). IB can be negative (meaning that the switched shunt is out of service) or positive (meaning that the switched shunt is in-service).

    4. MODE 6 part of the code can be modified to write out the new keyword ('USRSWS2') or the old keyword ('USRSWS') can be left as is. If the MODE 6 code is modified to write out the new keyword 'USRSWS2' then the switched shunt id (contained in the PSS®E data structure SWSHID(ISW)) will have to be written out. For details on the usage of 'USRSWS2' refer to Program Operation Manual (and search for USRSWS2). Note that the contents of the variable SWSHID is accessible to users via the INCLUDE of 'COMON4.INS' in the user model code.

    5. PSS®E Library models (i.e., models supplied as part of PSS®E installation) of switched shunts like ABBSVC1, CHSVCT, SWSHNT, CSSCST, SVSMO1T2, SVSMO2T2 have all been converted to the corresponding new model names ABBSVC2, CHSVCT2, SWSHNT2, CSSCST2, SVSMO1T2, SVSMO2T3 respectively. Dynamics data files (dyr files) with old model names need not be changed. When the dyr file contain-

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ing the old models names are read in, the DYR read operation in PSS®E will automatically convert these models to their new names with the model applied to the first switched shunt at that bus. If the dyr file uses the newer model names, then the user will have to specify a switched shunt id as part of the dyr record.

- Pre-version 35 dynamic snapshots containing PSS®E Library models will automatically work without any changes with the model attached to the first switched shunt at that bus.

7.5. Changes in User-Written Models

User-written models that use any of the functions - ADINTN, ADCHRB, ADCHRB, ADCHRN, ADDECN, will need to be modified. The changes required are described below in the example using ADINTN and apply to the other ADxxxxx functions as well.

Existing code:

```
EXTERNAL ADINTN
INTEGER IBPTR
...
CALL ADINTN(J, CB1, IBPTR)
```

The above code, after changes will look as follows:

```
INTEGER(KIND=C_INTPTR_T) IBPTR
...
CALL ADINTN(J, CB1, IBPTR)
```

In summary, the CALL to ADINTN still is the same, just that any EXTERNAL declaration for ADINTN should be removed, and also the INTEGER declaration for the variable corresponding to IBPTR in your code, needs to have a type declaration as shown above (using the KIND definition).

7.6. Program Corrections

The following issues have been identified and corrected in this release:

- When a single node within a substation becomes isolated due to the opening of a substation switching device, the potential for having the associated bus of that node erroneously taken out of service has been corrected.
- The node-breaker isolate function has been updated to correct behavior in situations where a winding of a three-winding transformer is impacted by the isolate process and the terminal bus connections at two or more of the winding ends of the three-winding transformer reside in the same substation.
- Bug fix in REECB1 and REECC1 models: In imposing the lower limit on STATE(K+5) (shown as s5 in the block diagram), the correct value of the lower limit was not being used.
- In WECCDS AC7B conversion, if Tdr (i.e., CON(J+4)) is zero in the “dyd”file, the conversion to “dyr” will be done by setting Kdr (i.e., CON(J+3)) to zero and Tdr to 1.0. This was done to prevent division by zero in handling of the derivative block.
- Corrected an error in SVSMO3T2 model dealing with the calculation of error signal (err) at initialization. Because of this bug, users would see a message (Warning: Initial error signal err 1.036 above CON(J+5)).

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- Fixed issues causing PSS®E to crash when performing interactive dynamic data checks in CIM5xx, CIM6xx and CIMWxx (xx could be BL, OW, ZN, AR or AL) models.
- Fixed issues causing PSS®E to crash when performing interactive dynamic data checks in the Maximum Excitation Limiter model MAXEX1.
- Fixed an issue in the load shed logic pertaining to frequency relay part of the load relay model UVUFBLU1. This bug was preventing load being shed by the frequency relay. Additionally this bug would have resulted in the model VAR location of another model getting changed.

  Fixed an issue with AC7C model in which the OEL and UEL model were not being handled correctly when these were placed at the summing junction.

  Fixed an issue with AC8C model in checking for OEL and UEL positions.
- Fixed issues in invoking the following dynamic simulation APIs:

  1. SET_GENANG_2
  2. SET_GENANG_3
  3. SET_GENANG_SUBSYS_FLAG
  4. SET_GENANG
  5. SET_GENPWR
  6. SET_GENSPDEV_SUBSYS_FLAG
  7. SET_GENSPDEV
  8. SET_RELANG
  9. SET_RELSCN_SUBSYS_FLAG
  10. SET_RELSCN
  11. SET_VLTSCN_SUBSYS_FLAG
  12. SET_VLTSCN
  13. SET_VOLT_VIOL_SUBSYS_FLAG
  14. SET_VOLTAGE_DIP_CHECK
  15. SET_VOLTAGE_REC_CHECK
  16. TRIG_VOLT_VIOLATION_CHECK
- Fixed numerous issues in AcccBrwsGrid when opening and browsing ACC files.
- Fixed an issue in ANSI when processing Induction machines.
- Fixed an issue when recording SPOUT_2.

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7.7. Known Issues

The following issues have been identified and known to exist in this release:

- All recent Intel Graphics drivers released since May 15, 2019 AND before August 12, 2020 appear to have issues which result in the incorrect display of slider diagrams. Do not update your Intel Graphics driver to anything above 26.20.100.6709 OR below 23.20.16.7584. If you have a version that is newer, please uninstall that version and install version 23.20.16.7584. Alternatively, if you have a discrete graphics card from AMD or Nvidia, make sure to configure your machine so that the discrete card is used for PSS®E instead of the integrated Intel processor. If you do not wish to update your drivers, you can also choose to turn off Hardware Acceleration by going to [Edit->Preferences...], [Diagram] tab, [Advanced Global Settings] tab. This will force the diagrams to use the Microsoft provided software-only rendering, but this is much slower than using a graphics processor, and much slower in Windows 10.
- When using a CodeMeter network license, the computer acting as the license manager will need the CodeMeter runtime upgraded to a minimum of 6.80. Download and install the latest version of the CodeMeter User Runtime from www.CodeMeter.com.
- Use of ACCC_PARALLEL, ACCC_PARALLEL_2, ACCC_WITH_TRIP_PARALLEL, N11_ACCC_PARALLEL, MACCC_PARALLEL, or MACCC_WITH_TRIP_PARALLEL are not permitted with a case containing substation data. This is being addressed and will be resolved in a future release.
- Currently it is not possible to query existing Report Tables using Python. This functionality will be added in a future release.
- Currently, only select activities use the Report Table format. The Report Table format will be rolled out to include additional activities in future releases.
- MUST as a Module reports are not currently written to Report Tables. They will be added in a future release.
- There are several known issues when switching back and forth repeatedly between MUST and PSS®E functionality in the same session. These issues are being addressed and will be fixed in future releases.
- Text containing non-ascii characters may produce an error in Python or may be represented incorrectly with Python data retrieval APIs. This issue will be resolved in a future release.
- Trying to open a case that contains either GNEs or GMBs will result in a crash caused by an x64 incompatibility. This will be fixed in the next release

7.8. Compiler Information

7.8.1. General Information

PSS®E-35 is built using the Intel® Parallel Studio XE 2017 Composer Edition (Intel®Visual Fortran, or IVF, 19.0) compiler and the Microsoft®Visual Studio 2017 (Microsoft®Visual C/C++ 2017) compiler. This is the recommended platform to use for building user-written extensions to PSS®E-35, such as CONEC, CONET, and user-written dynamics models. FORTRAN code must be compiled with IVF 12.X or later. VS2019 Community Edition can be downloaded from the Microsoft website for free and has been tested with the PSS®E Environment Manager. Intel® Parallel Studio comes with a Visual Studio Shell, which will work with PSS®E-35 as well. When using a version of IVF other than 19.0, please be aware that the later run-time libraries must be used. The PSS®E Environment Manager, which is delivered with PSS®E-35 or can be downloaded from our

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website, can be used to build extension libraries for PSS®E or assist with maintaining environment variables associated with controlling the compiler versions in use.

7.8.2. Additional Notes

A compiler is not required to run PSS®E because the Fortran and C run-time libraries are included with PSS®E. A compiler is required only if you are compiling CONEC and/or CONET, user-written dynamics models, or programs that call USRCAS.

Intel®has offered a special discounted upgrade fee to Siemens PTI customers. To purchase either the upgrade or a new license of the IVF compiler under this offer, you must contact Fred Fan at fred.fan@intel.com. Please be sure **Siemens PTI customer** is included in the subject line.

If you are still using CONEC/CONET type models, we recommend that you begin the process of converting them to table-driven models so that they may be distributed as DLLs and not require the use of a compiler by each end user. The support for CONEC/CONET models will be removed in a future release, so it is imperative that the migration process to move away from CONEC/CONET to table-driven begin as soon as possible.

7.9. Program Documentation

Documentation can be found in the DOCS folder in the PSS®E-35 installation directory.

7.10. Important Compatibility Notes

The following items address issues with backwards compatibility:

- PSS®E-35 SAV files are not compatible with previous versions of PSS®E.
- PSS®E-35 SNP files are not compatible with previous versions of PSS®E.
- PSS®E-35 SLD files are not compatible with previous versions of PSS®E.

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